

Chapter 24: Eigenvalues and eigenvectors Worksheet

1. Are each of the following statements true or false?

- (a) For arbitrary given matrix A , since $A\mathbf{0} = \mathbf{0}$, then A always has eigenvalue 1.
- (b) For arbitrary given square matrix A , A and A^T have the same eigenvalues.
- (c) A is an upper(lower) triangular matrix, then the eigenvalues of A are the elements on the main diagonal of A .
- (d) A is singular if and only if 0 is an eigenvalue of A .
- (e) If A is a 4×4 matrix, and 2 is an eigenvalue of A , then 8 is an eigenvalue of A^3 .

2. The eigenvalues of $A = \begin{bmatrix} -2 & 5 & 5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$ are -2 and 3 . Find a basis for the eigenspace corresponding to each eigenvalue.

3. Find all eigenvalues of each matrix A , and find a basis for the eigenspace corresponding to each eigenvalue.

(a) $A = \begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix}$

(b) $A = \begin{bmatrix} 4 & 2 \\ -2 & 8 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & -5 & 0 \\ 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$